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What can rocks tell us about the past?

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Abstract

Did you know the Grand Canyon can help us figure out what happened on Earth long ago? As in, 1.7 billion years ago? As we go down into the Grand Canyon, we find older and older rocks. Some of the oldest rocks are in a section of the canyon called the Upper Granite Gorge. These blocks of metamorphic rock formed when older rocks experienced extreme heat and pressure. We wanted to figure out the pressure experienced by four of the blocks. We collected five samples and analyzed them. We used the minerals in the

rocks to determine the pressure each block experienced. We learned that the blocks formed under different pressures. We also found out that the blocks that formed at higher temperatures experienced higher pressure than blocks that formed at lower temperatures. The difference in pressure made us realize the blocks formed at different depths. Then they moved when tectonic plates pushed together to form mountains.

Introduction

The Grand Canyon is famous for its natural beauty. Layers of rock make up the canyon's walls. These layers give the Grand Canyon its vibrant colors. Also, these same **rock layers give scientists a window into Earth's past.** As we go deeper into the canyon, we find older rocks that look different than those above. Analyzing these rocks tells us how they formed and how things changed over time.

Upper Granite Gorge is a part of the Grand Canyon. It contains rocks from more than 1.7 billion years ago. The section contains seven rock blocks. Some of the blocks seem to have formed under different conditions than the others. These rocks are **metamorphic**. That means they experienced extreme heat and pressure. Scientists think these metamorphic rocks formed during a major mountain-building event. During this event, the movement of **tectonic plates** caused a volcanic island chain to collide with the ancient continent of **Laurentia**. The volcanic island chain became a part of Laurentia.



A



B

(A) We went down into the Upper Granite Gorge in the Grand Canyon. (B) A photo of metamorphic rocks we found there.

The blocks that make up Upper Granite Gorge experienced different temperatures and pressures. We classify some of the blocks as hot blocks and some as cold blocks. Hot blocks formed at higher temperatures than cold blocks. We wanted to determine and compare the pressure experienced by the hot and cold blocks. Also, how could these blocks be near each other if they formed under different pressure conditions?

Methods

We collected five samples from different blocks in Upper Granite Gorge (Figure 1). Two samples came from the same cold block. Three samples came from hot blocks. Then we conducted two types of analyses.

1. Minerals

We identified the minerals that one sample from Cold Block 1 contained. We also identified the minerals that the Hot Block 1 sample contained. We shined high-energy **X-rays** on each sample. The minerals in the samples then released a unique set of X-rays. We analyzed the sets of X-rays to determine which minerals the samples contained. We also used a computer program to figure out the temperature and pressure needed for this group of minerals to form.

2. Garnets

We analyzed garnets from all five samples. Some of these garnets had quartz crystals trapped inside. We used a special probe to determine the composition of the garnets. Then we shined a green laser on the quartz crystals. The light **scattered** off these crystals. We determined the **wavelengths** of the scattered light. Then we compared these wavelengths to the wavelength of the laser. We used how much they differed to figure out the amount of pressure the rocks experienced.

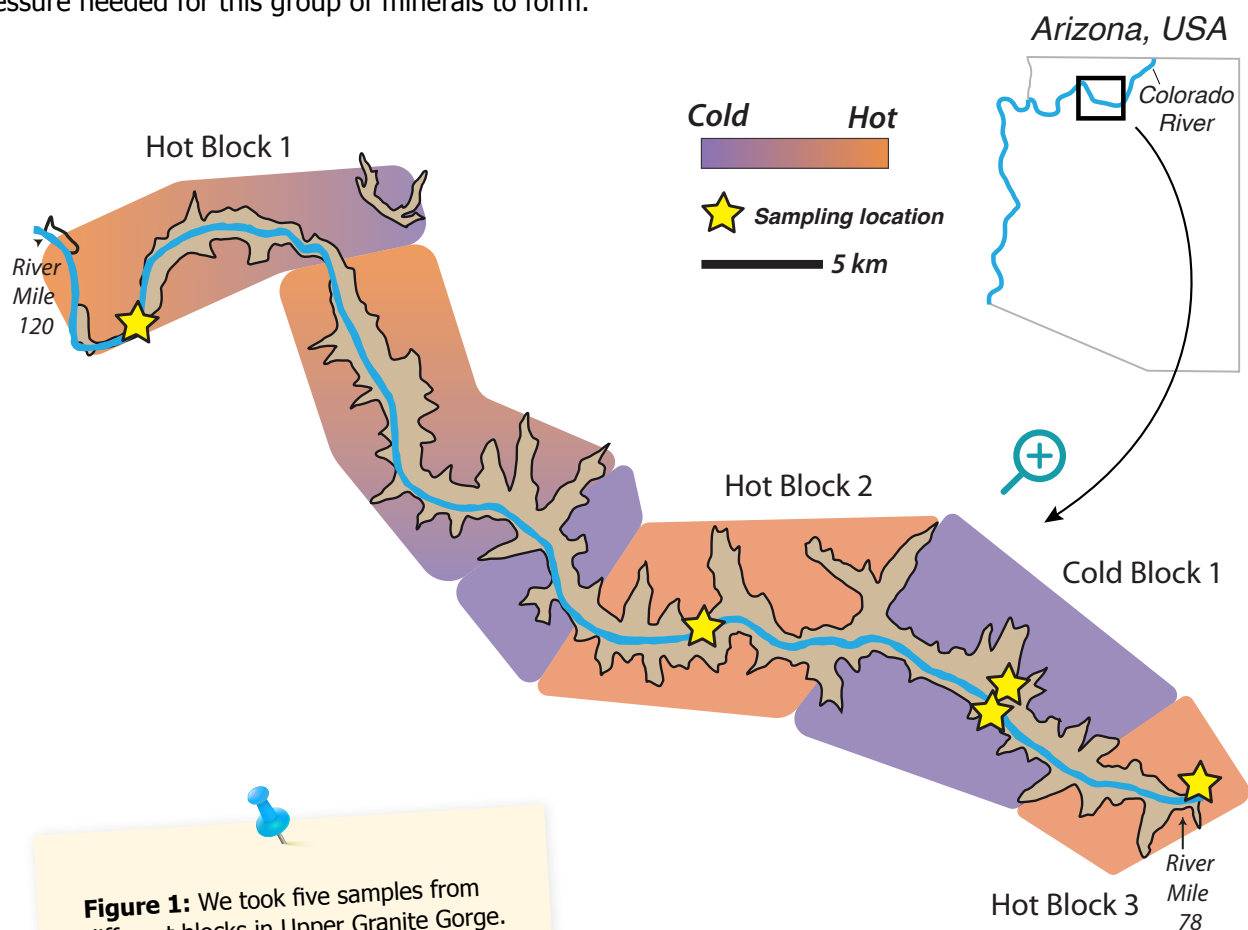


Figure 1: We took five samples from different blocks in Upper Granite Gorge.

Results

1. Minerals

The sample from Cold Block 1 contained seven minerals. Some of these minerals were garnet, staurolite, and quartz. The highest temperature that Cold Block 1 experienced was 625°C. It experienced low to medium pressure.

The sample from Hot Block 1 contained six minerals. Some of these minerals were garnet, cordierite, and quartz. It also contained partially melted minerals. This block experienced temperatures greater than 700°C. It experienced medium to high pressure.

2. Garnets

Garnets and trapped quartz crystals from Cold Block 1 experienced temperatures between 530°C and 555°C (Figure 2). The pressure was about 5 **kbars**. The samples from Hot Block 1 and Hot Block 3 were similar. The outer layers of the garnets formed between 720°C and 730°C. This part of the garnets formed at about 7 kbars. The garnets in Hot Block 2 formed at lower pressures or higher temperatures than the other two hot blocks.

How did the pressure experienced by the hot and cold blocks compare?

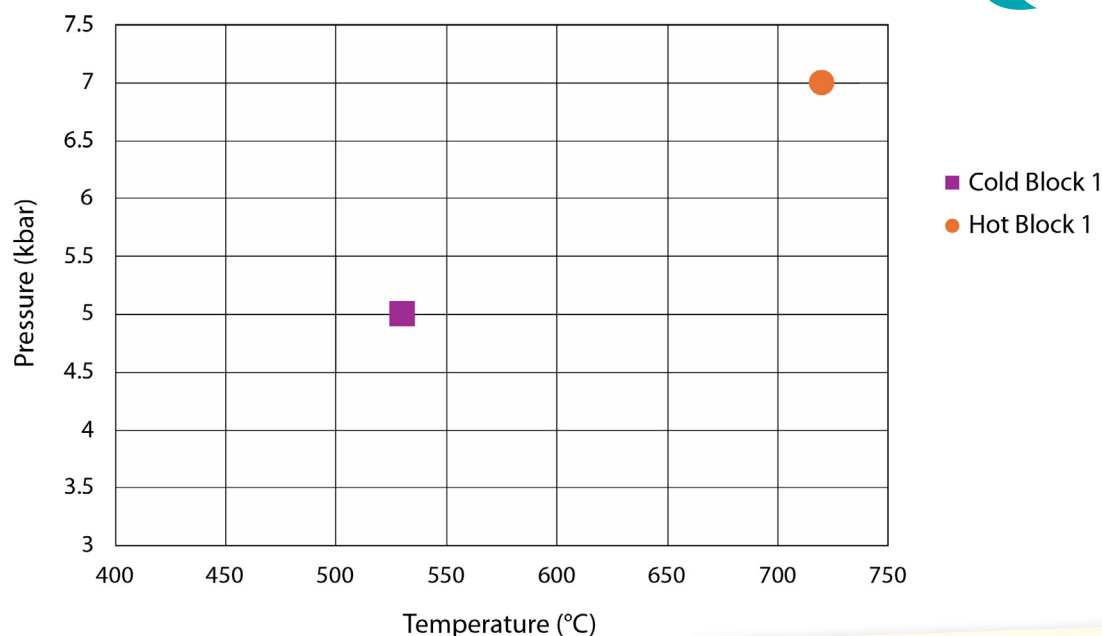


Figure 2: The estimated average pressure and temperature conditions for garnet samples from Cold Block 1 and Hot Block 1.

Discussion

We learned that the cold and hot blocks experienced different pressures. **Cold blocks formed under pressures that were about 2 kbars lower than the hot blocks.** This pressure difference tells us that the blocks were originally

located at different depths. The hot blocks were deeper, causing them to experience higher pressure. But if the blocks were at different depths, how are they next to each other today?

Our analysis supports the idea that the blocks moved after they became metamorphic. That means there were two mountain-building events that occurred. The first one caused the rocks in the blocks to form. The second one caused the blocks to move.

Before we conducted this research, scientists thought that

the blocks changed between 1.71 and 1.69 billion years ago. The pressure data we collected allowed us to adjust this time to an earlier range. Now we think the rocks changed between 1.75 and 1.70 billion years ago. This means that the data give a more detailed timeline for the area.

Conclusion

Studying the rocks in the Upper Granite Gorge of the Grand Canyon helped us adjust our thinking. As you explore the world around you, it is important to think like a scientist. Collect data to better understand what is happening and

why it is happening. Keep an open mind and adjust your thinking as you collect more evidence about an event or situation. Remember: just like rocks change, so does our understanding of the world!

Glossary of Key Terms

Kilobar (kbar) - a unit of pressure used to describe extremely high pressures.

Laurentia - an ancient continent that was made up of part of North America and Greenland.

Metamorphic - describes a type of rock that forms when an older rock experiences extreme heat and pressure. Examples include slate and marble.

Scattered - when light hits a surface or particle and goes in many different directions.

Tectonic plates - large pieces of solid rock that move and interact, causing the formation of features such as mountains and volcanoes.

Wavelength - the distance between two peaks or two troughs of a wave. This determines the color of light.

X-ray - a type of electromagnetic radiation with shorter wavelengths than visible light.

REFERENCES

Suzanne Autrey-Mulligan, Chloe Bonamici, Michael Williams, Karl Karlstrom, and Cailey B. Condit (2024) *Resolving Pressure Differences Within the Grand Canyon Precambrian Basement: Implications for Proterozoic Tectonics*. Geology.

<https://pubs.geoscienceworld.org/gsa/geology/article-abstract/52/4/266/633577/Resolving-pressure-differences-within-the-Grand>

National Park Service: Telling Time at Grand Canyon National Park

<https://www.nps.gov/articles/000/grcatime-vishnu-basement-rocks.htm>

Britannica Kids: Metamorphic Rock

<https://kids.britannica.com/students/article/metamorphic-rock/329895>

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Check your understanding

1

Why are many of the rocks in Upper Granite Gorge metamorphic?

2

How did the mineral composition of Hot Block 1 and Cold Block 1 compare?

3

How could the rock blocks be near each other even though they formed at different pressures?

4

With a partner, create a timeline of at least three events that occurred to form the rocks in Upper Granite Gorge. Include illustrations to provide a visual for what happened at each point in the timeline.

5

What are the mountains closest to where you live? Research how they formed. Write a summary to describe what happened when these mountains formed. How did this mountain-building event compare to the one that formed the rocks in Upper Granite Gorge?
